# Basic Inferential Data Analysis

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## Basic Inferential Data Analysis

We did a descriptive analysis. followed by confidence intervals and hypothesis testing.

### Descriptive Summary

```
knitr::opts_chunk$set(echo = TRUE)
library(summarytools)


data("ToothGrowth")
head(ToothGrowth)
```

```
##    len supp dose
## 1  4.2   VC  0.5
## 2 11.5   VC  0.5
## 3  7.3   VC  0.5
## 4  5.8   VC  0.5
## 5  6.4   VC  0.5
## 6 10.0   VC  0.5
```

```
dim(ToothGrowth)
```

```
## [1] 60  3
```

```
dfSummary(ToothGrowth)
```

```
## Data Frame Summary
## ToothGrowth
## Dimensions: 60 x 3
## Duplicates: 5
##
## ------------------------------------------------------------------------------------------------------
## No   Variable    Stats / Values           Freqs (% of Valid)   Graph           Valid      Missing
## ---- ----------- ------------------------ -------------------- --------------- ---------- ---------
## 1    len         Mean (sd) : 18.8 (7.6)   43 distinct values         : . :     60         0
##      [numeric]   min < med < max:                                  :   : : :     (100.0%)   (0.0%)
##                  4.2 < 19.2 < 33.9                               : . : : :
##                  IQR (CV) : 12.2 (0.4)                           : : : : :
##                                                                . : : : : : :
##
## 2    supp        1. OJ                    30 (50.0%)           IIIIIIIIII      60         0
##      [factor]    2. VC                    30 (50.0%)           IIIIIIIIII      (100.0%)   (0.0%)
##
```

```
## 3    dose        Mean (sd) : 1.2 (0.6)    0.50 : 20 (33.3%)    IIIIII          60         0
##      [numeric]   min < med < max:         1.00 : 20 (33.3%)    IIIIII          (100.0%)   (0.0%)
##                  0.5 < 1 < 2              2.00 : 20 (33.3%)    IIIIII
##                  IQR (CV) : 1.5 (0.5)
## ---------------------------------------------------------------------------------------------------
```

- len - Tooth Length - range from 4.2 to 33.9. mean 18.81 and median 19.25.
- supp - Supplement - OJ - Orange Juice - 30 observations and VC - Vitamin C - 30 observations.
- dose - Dose level - 3 doses - 0.5, 1, 2 mg/day for both OJ and VC.

## Overall Supplement Compare

```
t.test(len ~ supp, data = ToothGrowth)
```

```
##
##  Welch Two Sample t-test
##
## data:  len by supp
## t = 1.9153, df = 55.309, p-value = 0.06063
## alternative hypothesis: true difference in means between group OJ and group VC is not equal to 0
## 95 percent confidence interval:
##  -0.1710156  7.5710156
## sample estimates:
## mean in group OJ mean in group VC
##         20.66333         16.96333
```

Confidence Interval includes 0 [-0.171, 7.571] and p value greater than 0.05. There is no statistically significant difference in tooth growth between groups.

## Supplement at each 3 doses.

```
dose_0.5 <- subset(ToothGrowth, dose == 0.5)
dose_1.0 <- subset(ToothGrowth, dose == 1.0)
dose_2.0 <- subset(ToothGrowth, dose == 2.0)
```

For 0.5 mg/day

```
t.test(len ~ supp, data = dose_0.5)
```

```
##
##  Welch Two Sample t-test
##
## data:  len by supp
## t = 3.1697, df = 14.969, p-value = 0.006359
## alternative hypothesis: true difference in means between group OJ and group VC is not equal to 0
## 95 percent confidence interval:
##  1.719057 8.780943
## sample estimates:
## mean in group OJ mean in group VC
##            13.23             7.98
```

At 0.5mg/day, Orange Juice is more effective than Vitamin C.

For 1 mg/day

```
t.test(len ~ supp, data = dose_1.0)
```

```
##
##  Welch Two Sample t-test
##
## data:  len by supp
## t = 4.0328, df = 15.358, p-value = 0.001038
## alternative hypothesis: true difference in means between group OJ and group VC is not equal to 0
## 95 percent confidence interval:
##  2.802148 9.057852
## sample estimates:
## mean in group OJ mean in group VC
##            22.70            16.77
```

At 1mg/day, Orange Juice is more effective than Vitamin C.

For 2 mg/day

```
t.test(len ~ supp, data = dose_2.0)
```

```
##
##  Welch Two Sample t-test
##
## data:  len by supp
## t = -0.046136, df = 14.04, p-value = 0.9639
## alternative hypothesis: true difference in means between group OJ and group VC is not equal to 0
## 95 percent confidence interval:
##  -3.79807  3.63807
## sample estimates:
## mean in group OJ mean in group VC
##            26.06            26.14
```

At 2mg/day, no siginificant difference seen.

## Conclusions and Assumptions

Orange Juice deliver tooth growth over vitamin C except in 2mg dose.